

Certified analytical bounds for hypersonic reachable sets

A measure-theoretic profile decomposition in validated interval arithmetic

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Abstract

The reachable set of a nonlinear control system can be embedded in a space of occupation measures, where the dynamics become a linear constraint and the reachable set is the union of the supports of the admissible terminal measures. This research asks what that embedding yields in a singular limit, for a vector field known only as a set-valued map enclosed in validated interval arithmetic. On the state space nothing is lost—the family is tight—but its weak limit forgets every event that lasts a vanishing time, and a reachable set is a support, not an integral: the limit satisfies the Liouville equation of the limit dynamics only up to a defect. The target theorem identifies that defect. The loss is read on the energy dissipation along the rescaled time axis, where Lions’ concentration-compactness alternative applies as it stands, and it organizes into a profile decomposition: a limit system in a few slow coordinates, one boundary-layer bubble per concentrating event, entering as a jump, and a remainder. The statement aimed at holds at fixed ε : the reachable set lies in that of the limit hybrid system, thickened by a ball whose radius is an explicit function of ε . Compactness gives the structure and no constants. Those are to come from direct estimates—the energy identity, a matching estimate at each pass, and averaging over the faster oscillations, where frequencies that cross are the critical difficulty—and are required in closed form, in the physical constraint set and the aerodynamic interval widths. Hypersonic entry supplies the instance: a six-degree-of-freedom vehicle whose thin-atmosphere limit furnishes the singular perturbation, whose exo-atmospheric arcs and equilibrium glide furnish the limit system, and whose atmospheric passes furnish the bubbles. The aim is a bound whose dependence on the data is explicit: valid uniformly over the entire admissible family, in the minimal physical state space, and re-evaluated rather than re-solved when the vehicle or the corridor changes.

1 The analytical problem

The object to be bounded is the reachable set of an atmospheric entry vehicle—every state it can be driven to under admissible control—which is what safety and threat analyses need enclosed. The flight-mechanics specifics are deferred: [Section 3](#) states why entry is the right instance and what its state space looks like, and the coupled dynamics—frames, state vector, translational and rotational equations, the aerodynamic bridge from attitude to force, and the glide condition—are carried, together with sketches of the limit argument and of the constants it rests on, in companion appendices omitted from this copy. What matters first is that the problem is an analytical one before it is an aerospace one, and that the existing methods fall short of it on two independent counts: the dimension of the state, taken up with the application in [Section 3](#), and the provenance of the guarantee, in [Section 1.2](#).

1.1 Reachable sets as occupation measures

Embed the flight dynamics in a space of measures. Along a trajectory of $\dot{x} = f(x, c, t)$, with the state x in the corridor $\mathcal{X}$ and the control c in $\mathcal{U}$, the time spent in each set defines an occupation measure μ on $[0, T] \times \mathcal{X} \times \mathcal{U}$, and the nonlinear flight mechanics become *linear*: the Liouville equation is a linear constraint tying μ to an initial measure and to a terminal measure μ_T , and the reachable set at time T is the union of $\text{supp } \mu_T$ over the measures that satisfy it. No single measure bounds it. The outer bound comes from the dual: a pair of functions (Φ, Ψ) , the first non-decreasing along every admissible trajectory and at least one on the initial set, the second majorizing it at time T , encloses the reachable set in the superlevel set $\{\Psi \geq 1\}$. This is the occupation-measure program of Lasserre, Henrion and collaborators [\[1, 2\]](#), developed for regions of attraction and invariant sets [\[3, 4\]](#) and, for hybrid systems, for reachable sets and optimal control, where a measure on each guard carries the mass from one mode into the next [\[5, 6\]](#). The embedding buys linearity at the price of a relaxation, and the price is known: a measure that satisfies the Liouville constraint is generated by a family of trajectories of the *convexified* inclusion $\dot{x} \in \text{conv } f(x, \mathcal{U}, t)$, not by a single trajectory of the original system [\[3, Lemma 3\]](#). The relaxed reachable set contains the true one, so an outer bound on the first is an outer bound on the second, which is the direction needed here.

The same relaxation admits the interval envelope. Because the true aerodynamics are transcendental and known only to within an envelope, they are enclosed in intervals, and a vector field selected from an interval is a set-valued map; the resulting differential inclusion is the object the measure must describe, not the nominal field [\[7, 8\]](#). Written with the selection as an additional box-valued input, the inclusion is a control system with a bounded disturbance, and at fixed ε its Liouville constraint is the standard one. What is not standard is carrying the outer approximation through the singular limit of [Section 2](#), uniformly over the selections. Separating time scales is itself a standard device of flight mechanics [\[9\]](#), and the classical reduction, which sets $\varepsilon = 0$, requires the fast motion to settle. Occupation-measure theory for inclusions that does not require it already exists, and it is the nearest prior art: it defines an occupational invariant measure without assuming solution uniqueness, which is the same move made here, and lets the slow dynamics be driven by the invariant measures of the fast [\[10\]](#). What it assumes is worth stating plainly, because it locates where the entry problem leaves it. The linear growth condition on the field serves only to exclude finite-time blow-up, and is automatic on a bounded corridor. The singular-perturbation theorem, however, assumes a uniform bound on the fast state along the whole ε -family [\[10, 11\]](#), and the natural fast state of entry is the altitude measured in scale heights: on a corridor of fixed altitude extent it ranges over an interval of length proportional to $1/\varepsilon$, the aerodynamic terms grow exponentially in it, and a skip that rises through a fixed altitude carries it out of every bounded set.

The hypothesis holds for motion that stays within a bounded number of scale heights of the glide, and it fails exactly where a pass concentrates; [Section 2](#) says which is which.

Objective 1: Interval embedding, uniform through the singular limit

Question. What do the occupation measures of the interval inclusion $\dot{x} \in \mathcal{F}_\varepsilon(x, c, t)$ converge to as $\varepsilon \rightarrow 0$, and what linear constraint does the limit satisfy?

To establish. At fixed ε the inclusion is a control system with a box-valued disturbance and its Liouville constraint is standard. So are the bounds that hold uniformly in ε : on mass, on support and, through the energy balance, on dissipation. Two things are not. The first is a certified upper and lower envelope of the atmospheric density, which fixes $\mathcal{F}_\varepsilon$ concretely. The second is the limit of the constraint itself. A weak-* limit of the measures satisfies the Liouville equation of the limit flows only up to a defect, carried where the dissipation concentrates. The claim to be proved is that this defect is a jump term, on a measure supported on the graph of the pass map, so that the limit satisfies the Liouville equation of the hybrid system of [Section 2.4](#), in the form with guard measures [\[5, 6\]](#), uniformly over the selections. Where nothing concentrates the limit is of the kind the existing theory describes, with the oscillation about the glide represented by an invariant measure [\[10\]](#).

Success. The limit constraint stated for the full inclusion, with each hypothesis either verified on the entry corridor or named as an assumption; and its dual, a pair (Φ, Ψ) non-decreasing along the flows and across the jumps of the limit system, as the form in which the outer bound of [Section 1.1](#) survives the limit.

1.2 The provenance of the guarantee

A polynomial barrier whose coefficients come from an interior-point solver has analytical *form* but inherits the solver’s accuracy [\[12\]](#), and the documented failure modes of numerical semidefinite solvers include the loss of strict feasibility on which the repair schemes depend [\[13\]](#). A solver’s output can be made into a certificate: rounded to rational coefficients and re-verified exactly, it is a proof [\[14\]](#). The obstacle is scale, not principle. The rounding needs a strictly feasible Gram matrix, which a tight bound does not have, and exact rational certificates carry a documented bit-complexity whose exponent grows with the number of variables [\[15, 16\]](#)—a cost the thirteen-state field of [Section 3](#) does not invite as a main line.

The program therefore fixes an order of evidence. The certificate itself is a closed-form bound: the constants of [Section 2](#), derived from the decomposition and evaluated over the corridor in validated interval arithmetic. Behind it stand symbolic checks of the derivations those constants rest on, run as part of the build against the same definition of the dynamics that is evaluated numerically. A moment-sum-of-squares computation on a reduced model is the third tier: an independent cross-check, to which the exact-certification machinery above belongs, and never the source of the guarantee.

Interval arithmetic is the setting in which a guarantee has meant, for forty years, an enclosure rather than an approximation. It is rigorous in floating point: every operation is rounded outward, so the computed interval contains the exact result, and nothing is gained for soundness by exact rational arithmetic, which cannot in any case evaluate an exponential or a sine [\[17, Ch. 3\]](#). Two distinct losses have to be managed, and they are often run together. The first is the *dependency problem*. Interval arithmetic is inclusion isotonic—the property that makes an enclosure valid at all—but not every law of real arithmetic survives the move to intervals: distributivity weakens to subdistributivity, and repeated occurrences of a variable are treated as independent [\[17, Chs. 4–5\]](#). The second is the *wrapping effect*: the image of a box under a flow is not a box, and re-enclosing it

at every step admits points that no trajectory reaches [17, Ch. 10]. Refining the step does not help. Under interval Euler steps of size h , re-boxed at each, a rigid rotation inflates a box by $(1 + h)^{2\pi/h}$ per revolution, which *increases* to $e^{2\pi} \approx 535$ as $h \rightarrow 0$; what controls wrapping is the representation of the set—Lohner’s QR factorization, zonotopes, Taylor models [18, 19]. Applied interval analysis supplies the machinery the pipeline is built from, and it is deliberately conservative: certified set inversion, contractors, and validated integration [20].

Objective 2: Validated computational pipeline

Question. Can the closed-form constants, the pass enclosures they do not cover, and the reachable set of the limit system of Section 2.4 be evaluated over the whole corridor with every rounding error accounted for?

To build. An interval evaluator for the transcendental field and its Jacobian with outward rounding, contractors for the corridor constraints, and a validated integrator for the passes, extending the ball-arithmetic prototype described in Section 4.1. The limit system is itself a reachability problem, in the few slow coordinates and over every bank history: it is to be enclosed by comparison bounds in closed form where they exist—the certified range bound of Section 4.1 is the first—and by validated set propagation where they do not.

Success. Every inequality the certificate asserts is discharged by a program that returns either a proof or an explicit failure to decide at the working precision, and the dynamics it evaluates are the dynamics that were checked symbolically.

2 The singular limit and its profile decomposition

This section carries the new content. The instance supplies a small parameter—in the entry problem the thin-atmosphere ratio $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$, where H_s is the atmospheric scale height and R_e the planetary radius, written ε where the other small parameters are not in play—and the question is what the occupation-measure family does in that limit.

2.1 Loss of compactness along the time axis and the defect measure

On the corridor nothing is lost in the usual sense. The set $[0, T] \times \mathcal{X} \times \mathcal{U}$ is compact, so the occupation measures—which live in the cone $\mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U})$ of nonnegative Radon measures, topologized weakly-* as the dual of $C([0, T] \times \mathcal{X} \times \mathcal{U})$ —form a tight family and converge along subsequences. The limit is nevertheless the wrong object, because it forgets the atmospheric passes. How much it forgets depends on which limit is taken, and that has to be said first.

A pass entered at a flight-path angle γ_e turns the velocity through about $2\gamma_e$ and sheds a share of the kinetic energy of order $\gamma_e/(L/D)$, since lift cannot be had without drag. It crosses a scale height in a time $H_s/(V\gamma_e)$, of order $\varepsilon_{\text{atm}}/\gamma_e$ against the slow time, and it levels out where the lift is $V^2\gamma_e^2/(2H_s)$, of order $\gamma_e^2/\varepsilon_{\text{atm}}$ against gravity. The entry angle therefore enters through $\gamma_e/\sqrt{\varepsilon_{\text{atm}}}$ —as in the classical Chapman–Loh variables, it is $\sqrt{\varepsilon_{\text{atm}}}$ and not ε_{atm} that sets the scale [21, 22]—and two limits have to be told apart. With $\gamma_e/\sqrt{\varepsilon_{\text{atm}}}$ held fixed, a pass lasts a time of order $\sqrt{\varepsilon_{\text{atm}}}$, sheds an energy of the same order, and loads the vehicle by a fixed multiple of what the glide does. This is the limit of a corridor whose dynamic-pressure and load limits are held fixed, and in it nothing concentrates: the passes are many and slight, the troughs of an oscillation about the glide. With γ_e held fixed, a pass lasts a time of order ε_{atm} and sheds a fixed share of the energy, under loads that grow like $1/\varepsilon_{\text{atm}}$. It carries occupation mass of that order, and none in the limit, while its share of what distinguishes one trajectory from another—the energy dissipated, the heat

absorbed, the turn of the velocity vector—does not vanish with it. The physical case lies between the two. $\sqrt{\varepsilon_{\text{atm}}} \approx 0.034$ is about two degrees and entry angles are a few degrees, and a computed pass is neither short against the arc that follows it nor distinct in scale from the oscillation about the glide. The statement aimed at is therefore one at fixed ε , with a threshold that decides which passes are resolved one by one, and the two limits are the ends it has to cover.

A reachable set is a support, not an integral, and what has no mass still counts. That alone does not set this problem apart from the limit theory of singularly perturbed control, whose value functions already carry a terminal cost on the slow state [11]. What sets it apart is the hypothesis under which that theory works: the fast state, and the trajectories that approximate an optimum, stay in a bounded set, and a pass that concentrates is what that excludes. The loss has to be read where it occurs, along the fast-time axis $\tau = t/\sqrt{\varepsilon_{\text{atm}}}$, on which the horizon $[0, T/\sqrt{\varepsilon_{\text{atm}}}]$ exhausts the half-line and translations act non-compactly.

Between passes the vehicle is outside the sensible atmosphere, on an arc governed by gravity alone; once captured, it stays near the equilibrium-glide balance $\Lambda = 1$. In the 3-DOF (point-mass) approximation this is the scalar condition $L \cos \sigma / [m(g - V^2/r)] = 1$; in the full 6-DOF formulation the lift component $L \cos \sigma$ is not an independent input but is determined by the attitude quaternion through the body-to-LVLH rotation, so the balance is a single scalar constraint on the 13-dimensional translational-rotational state, and the slow set is its intersection with the trim and quasi-static-deflection conditions of the two faster groups—one constraint set for each. The balance is taken with planetary rotation and oblate gravity retained, and the two do not enter on equal terms: flying east at entry speed the Coriolis term is tens of percent of the net acceleration $g - V^2/r$ that the lift has to balance, while the J_2 terms are a percent of it at most. The scalar $\Lambda = 1$ is the reduction under spherical gravity, no rotation and small flight-path angle.

The loss is read on a *defect measure*, and the object has to be chosen so that the passes carry mass. The part of the occupation-measure limit that lies off the glide cannot serve, since the passes have none. The energy dissipation can. Along every trajectory of the inclusion the specific mechanical energy in the rotating frame, $\mathcal{E} = \frac{1}{2}V^2 + \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$, changes at the rate $\dot{\mathcal{E}} = -P$, where $P = DV/m \geq 0$ is the drag power: neither the lift nor the Coriolis force does work. The total dissipation $\int_0^T P dt \leq \mathcal{E}_{\text{max}} - \mathcal{E}_{\text{min}}$ is therefore bounded uniformly over controls, over selections of the interval field, and over ε . Let

$$\nu_\varepsilon(\tau) = \frac{\sqrt{\varepsilon} P(\sqrt{\varepsilon} \tau)}{\int_0^T P dt} \quad (2.1)$$

be the dissipation density on the fast-time axis, normalized to unit mass and extended by zero. It is a defect measure in the working sense. Where it concentrates is where the limit of the occupation measures stops satisfying the Liouville equation of the limit flow, and the mass it puts there bounds the size of the failure, because the aerodynamic impulse over any interval is bounded by the energy dissipated in it. Along any sequence $\varepsilon_n \rightarrow 0$, with a trajectory chosen for each, the ν_ε are nonnegative densities of fixed mass on the line, which is exactly the hypothesis of Lions' concentration-compactness lemma [23, Lemma I.1], and here all three of its alternatives mean something:

- *vanishing*—no window of fixed fast-time width holds a fixed share of the dissipation: the glide and the oscillation about it, on which the drag is bounded and such a window holds a share of order $\sqrt{\varepsilon}$;
- *compactness*, up to translation—one pass carries the dissipation;
- *dichotomy*—the dissipation splits between pieces whose fast-time separation diverges: two passes a positive time apart, or a pass followed by a glide.

Which of the three occurs is decided by the scaling above. With the loads held fixed the drag power is bounded and every sequence vanishes. Concentration needs the fixed entry angle, and then translations along the fast-time axis separate the passes, and it is the dilation half of the principle—the limit-case modification [24]—that resolves each one at its own width, since a pass shorter than $\sqrt{\varepsilon}$ is a Dirac mass at that scale. The trichotomy is Lions’ lemma applied as stated. That the vanishing part is the glide and the concentrated pieces are the passes is asymptotics still to be argued, from the inner and outer expansions of the classical theory [25], and one step of the argument is open.

2.2 Skeleton, bubbles, residual

Rather than restore compactness, the program extracts the profile decomposition that iterating the dichotomy hands it:

- a *skeleton*, low-dimensional—what survives the limit: the exo-atmospheric arcs between passes, which are conics up to the oblateness correction, followed by the equilibrium glide, a graph over (r, V) in the corridor subspace once the balance is solved for the vertical lift fraction. For the glide a rigorous characterization with quantitative accuracy estimates already exists [26], and it treats the glide as a *regular* perturbation: the exact solution is not attracted to the zeroth-order one, a point that returns in [Section 2.4](#);
- one *boundary-layer bubble* per atmospheric pass, the analogue of a concentrating profile: the pass rescaled to its own width. Its net effect on the slow coordinates is a jump. The exit states that the bank history and the selections can produce from one entry state form a compact set, and the map that assigns it, the *pass map* J , is the object to be enclosed. Orthogonality is read off the scale-and-core separation condition that governs concentrating profiles generally [27]. The decomposition of skip entry into an aerodynamic inner region and a gravitational outer one is classical [21, 25], and it has been carried to second order and to three dimensions [28]. Those are expansions of one trajectory, with the error assessed by example. What is not classical is a statement uniform over a family of trajectories, with an explicit remainder at fixed ε ;
- a *residual*: what the finitely many profiles do not capture. In a profile decomposition the remainder is not compact and is not small in the conserved norm; it is small in a weaker one, and only as the number of extracted profiles grows [29, 30]. Here the weaker norm is the largest share of dissipation held by any fast-time window: on a glide the residual carries most of the dissipation, and almost none of it in any one window. The quantitative bound the program needs is a different statement, a distance in state space, and it is taken up in [Section 2.4](#).

The two kinds of statement are kept apart, because they come from different arguments. A decomposition obtained by compactness is a statement about subsequences, and what it gives uniformly it gives by contradiction: it says which pieces can occur, and that nothing else can, and it produces no constant. The wave-equation theory is explicit about this. The uniform bound that follows from its decomposition comes with a constant its authors say they cannot describe [29, Corollary 2]. Here compactness is asked only for the structure. The constants come from direct estimates: the energy identity, the bound it gives on the aerodynamic impulse, a matching estimate at each pass, and averaging about the glide. Each of those is an inequality with a constant that can be written down, and writing them down is objective 5.

The number of bubbles is bounded *a priori*, and the bound comes from the same monotone quantity. A pass that dissipates at least δ of specific energy can occur at most

$$K_\delta \leq \frac{\mathcal{E}_{\max} - \mathcal{E}_{\min}}{\delta} \quad (2.2)$$

times, for every control and every selection of the interval field. Passes below the threshold are left to the oscillation about the glide and to the residual, which is the device of the profile-decomposition literature: only finitely many profiles exceed any given size. The integrated heat-load budget gives the count its physical reading—a trajectory can only afford so many skips [25, 31]—but it routes through a stagnation-point heating correlation, and the energy budget does not.

Enclosing a pass is where the wrapping effect has to be met. For linear systems it can be avoided altogether: support-function methods propagate each direction independently, so successive over-approximations do not compound, and what remains is a discretization error of first order in the time step [32]. Nothing comparable is available for a nonlinear pass.

Question. Can the net effect of one atmospheric pass on the slow coordinates be enclosed tightly enough that a sequence of K_δ passes does not exhaust the corridor?

To prove and build. Wrapping in a pass has a specific source. Trajectories of a set enter the layer at different times, the set is sheared along the flow, and re-enclosing the sheared set in a box at every step admits points that no trajectory reaches. The remedy that fits the decomposition is to enclose the *pass map*—entry state to exit state, in the rescaled time of the profile—as a single object, a zonotope or Taylor-model enclosure of a map rather than a tube of boxes, and to compose pass maps with the outer flow.

Success. A bound on the growth of the enclosure per pass, explicit in the interval widths, and its verification on a multi-pass corridor against sampled trajectories.

2.3 Almost-orthogonality of the scaling groups

Three scaling groups act on the time axis. The atmospheric one dilates by $\sqrt{\varepsilon_{\text{atm}}}$; those of the attitude and the elastic motion dilate by $\varepsilon_{\text{att}} = T_{\text{att}}/T_{\text{slow}}$ and $\varepsilon_{\text{ela}} = T_{\text{ela}}/T_{\text{slow}}$, the short-period attitude and first structural periods measured against the slow time $T_{\text{slow}} = \sqrt{R_e/g}$. Their separation is the Bahouri–Gérard almost-orthogonality condition, read on the dissipation: two profiles are orthogonal when their scales diverge or, at equal scale, when their centers separate by a diverging number of widths [29, (1.18)]. Two things follow from it, and they differ in kind. The dissipation is a measure, so it splits *additively* across orthogonal profiles; that is the analogue of the energy identity of the wave-equation theory, and it costs nothing. The state does not split additively, because the profiles interact through the nonlinear dynamics: the state leaving one pass is the datum of the next arc. What has to be proved is the analogue of the nonlinear statement of that theory, that the contributions of separate profiles compose without interacting, up to an error that is explicit and uniform over the family. For passes separated in time this is a matching estimate between inner and outer solutions. For the three groups it is scale separation where the timescale bands are ordered; where two bands cross inside the corridor the groups are not orthogonal there, and the decomposition has to carry one joint profile on the coupled subsystem rather than two. Establishing this with the scales themselves known only to within the aerodynamic intervals is the one genuinely new obligation, and it sits on the critical path.

Objective 4: Almost-orthogonality under interval disturbances

Question. Do the scaling groups remain orthogonal, in the sense of the Bahouri–Gérard condition, for every selection of the aerodynamic interval field—and where they do not, what replaces the decomposition?

To prove. A non-interaction theorem: for profiles separated in scale or in time, the map from entry data to exit data factors through the individual pass maps and the outer flow, with an explicit remainder, uniformly over selections. The interval widths enter twice: as a disturbance within each profile, and through the periods T_{att} and T_{ela} , which depend on the uncertain coefficients, so that the ordering of the bands is itself a hypothesis to be verified over the corridor. Where two bands cross, the statement is for a joint profile.

Success. The theorem for the atmospheric and attitude groups with the remainder in closed form; the elastic group either included or enclosed as a widened interval, as in the fallback of Section 4.

The argument worth transporting is the constructive one: splitting a bounded sequence into an oscillatory and a singular part at each scale forces those scales to be orthogonal, and iterating extracts a pairwise orthogonal family, so orthogonality is produced by the construction rather than assumed by it [29].

2.4 The certified bound

The target statement is

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\left(\Phi_S(\Theta) \right)}_{\text{skeleton}} \oplus \underbrace{\left(\bigoplus_{j=1}^{K_\delta} \mathcal{I}_j \right)}_{\text{bubbles}} \oplus \underbrace{\mathbb{B}_\varrho(C_{\mathcal{R}}\sqrt{\varepsilon})}_{\text{residual}}, \quad (2.3)$$

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory. Here $\mathcal{R}_\varepsilon(T)$ is the set of slow states reachable at time T from the initial set, under every admissible control and every selection of the interval field. Θ is the compact set of skeleton parameters—initial data and slow controls—and $\Phi_S: \Theta \rightarrow \mathcal{X}$ the endpoint map of the skeleton flow. $\mathcal{I}_j$ is the set of displacements of the time- T endpoint attributable to pass j : its impulse set, carried to time T by the skeleton flow, over all states from which the pass can be entered. The sum $\oplus$ is the Minkowski sum, and $\mathbb{B}_\varrho(a)$ is the ball of radius a in the corridor metric ϱ , a weighted supremum norm in which altitude is measured in scale heights and each remaining slow coordinate in units of its corridor width. Passes compose in sequence, not in parallel, and the sum is the telescoped form of that composition: the endpoint of a trajectory with K passes differs from the skeleton endpoint by a sum of K displacements, one from each $\mathcal{I}_j$. The sum forgets that successive displacements are correlated, which makes the bound conservative and not unsound.

The residual constant $C_{\mathcal{R}}$ and the bubble count K_δ are to be derived as explicit, closed-form expressions in the corridor bounds and the aerodynamic interval widths, independent of any numerical optimization. Both exist in first form. The count is Eq. (2.2). On the glide, with L/D the lift-to-drag ratio and σ the bank angle,

$$C_{\mathcal{R}} = \frac{2\sqrt{gR_e}}{V(L/D)\cos\sigma\sqrt{1-V^2/(gr)}}, \quad (2.4)$$

which bounds the computed distance to the glide to within a few percent over the range of lift-to-drag ratio, speed and bank tested. The same analysis shows what the formula does not yet cover, and the

limitation is structural. $C_{\mathcal{R}}\sqrt{\varepsilon}$ is twice the damping ratio of the phugoid, the oscillation of altitude against flight-path angle about the glide. Its damping *rate*, $g/[V(L/D)\cos\sigma]$, does not depend on ε , while its frequency grows like $\varepsilon^{-1/2}$. In the limit the motion normal to the glide is an undamped oscillation on its own timescale: the glide balance is not a normally hyperbolic slow manifold in the sense the geometric theory requires [33], and no boundary-layer estimate of Tikhonov type brings a trajectory onto it. The constant above is the ringing amplitude of a trajectory started level on the glide; a trajectory leaving a pass rings with whatever amplitude the pass left it, for a time comparable to the entry. The residual bound therefore has to carry that amplitude, either as an explicit decaying term or by promoting the phugoid amplitude to a slow coordinate of the skeleton and averaging over its phase; and $C_{\mathcal{R}}$ diverges as V approaches circular speed, which is where the glide gives way to arcs and passes.

In the spherical-LVLH coordinates the altitude and speed bounds of the corridor are box constraints on (r, V) . The heating, dynamic-pressure and load limits are not: each is a curve $h \geq h_{\min}(V)$. Each is, however, monotone in altitude and in speed at fixed attitude, so contracting a box against it returns the exact interval hull of the feasible part, and the interval widths that enter $C_{\mathcal{R}}$ and K_{δ} inherit that tightness. Requiring the constants in closed form, rather than as the output of a solver, is what separates a certificate from a computation in the sense of Section 1.2.

Question. What are $C_{\mathcal{R}}$ and K_{δ} as functions of the corridor bounds, the initial set and the aerodynamic interval widths?

To derive. The count from the energy budget, with the threshold δ chosen against the residual it leaves behind. The residual in three parts: the lag behind the glide, which Eq. (2.4) already gives; the ringing left by a pass, which decays at a rate independent of ε and has to be carried explicitly or absorbed into the skeleton by averaging; and the matching error at each pass. Each constant is to hold for every selection of the interval field, and its range of validity—bounded away from circular speed—is part of the statement.

Success. Formulas whose evaluation over the corridor in validated interval arithmetic bounds the distance from sampled trajectories of the full model to skeleton plus bubbles, with no fitted number.

3 The application: hypersonic entry

Entry is not an arbitrary illustration of the machinery above. It supplies the singular perturbation, the skeleton and the concentrating events in one system, and it is a problem the existing methods are known not to reach.

3.1 Why existing methods do not reach it

The direct method solves a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in state dimension and capped in practice near four or five states [34, 35]. A rigid-body entry vehicle carries thirteen states,

$$x = (r, \lambda, \phi, V, \gamma, \psi, q_0, q_1, q_2, q_3, p, q_b, r_b) \in \mathbb{R}^{13}, \quad (3.1)$$

and $17 + 2N_m$ once integrated heat load, wall temperature, ablation recession, mass loss and N_m aerothermoelastic modes are coupled [36]. The two counts use “degree of freedom” in its two customary senses, and it is worth separating them: the vehicle is six-degree-of-freedom in the

configuration sense (three translational, three rotational), while Eq. (3.1) has thirteen coordinates spanning a twelve-dimensional state manifold once the unit-quaternion constraint $\|\vec{q}\| = 1$ is imposed.

So the grid method stops before the real problem starts.

Set-propagation methods go further. Taylor-model and zonotope flowpipes return rigorous outer bounds at ten to twenty states, and they are the baseline this program has to be measured against [19, 37, 38]. What they return is an enclosure of one instance: a numerical set for one vehicle, one initial box and one horizon, whose tightness tends to degrade with the width of the initial set and with the length of weakly damped motion it has to follow—and the phugoid is weakly damped for the whole of the glide. They are the right tool inside a pass, where the horizon is short, and that is how objective 3 uses them. They do not produce a bound whose dependence on the corridor and on the interval widths is explicit, which is what a statement uniform over a family requires.

What makes the measure-theoretic route viable is not that it evades the dimension but that the bound it produces does not scale with it: the skeleton is low-dimensional by construction, the bubbles are compact, and the residual is a single ball.

3.2 The state space and the corridor

The translational state is carried in geocentric spherical position (r, λ, ϕ) and wind-frame velocity (V, γ, ψ) , so that the altitude and speed bounds of the entry corridor are box constraints on states and the heating, dynamic-pressure and load limits are monotone in the two coordinates they bind—the natural setting for interval contraction. Attitude is a unit quaternion, free of Euler-angle singularities. The chart has a price, which is stated rather than hidden: the longitude and heading rates are singular at the geographic poles, so the corridor is taken with $|\phi| \leq \phi_{\max} < \pi/2$, and a trajectory class that overflies a pole needs a second chart or a Cartesian formulation.

Because interval arithmetic evaluates transcendentals natively, the algebraic adjunctions that recast the system in polynomial form [39] are not needed on the main line; they survive only inside the moment-sum-of-squares cross-check. The state space therefore carries only the physical dimensions—spherical position, wind-frame velocity, attitude quaternion, body rates, thermal, ablation, and modal states—with no artificial embeddings, and the two auxiliary states of the polynomial formulation, the density root $y = \sqrt{\rho/\rho_0}$ and the wall-temperature surrogate, are gone. This is a design decision rather than an open problem, and its trade is stated once: transcendental dynamics in coordinates native to the corridor and the glide, against polynomial dynamics in which the corridor is a nonlinear function of the state.

Abstract Independent Verification

Question. Is the certificate right, by evidence that does not share its derivation?

To build. Three checks, in the order of Section 1.2. Symbolic verification of every derivation behind the constants, run with the build. A moment-sum-of-squares outer approximation on a reduced polynomial model, as a cross-check that shares neither the decomposition nor the interval arithmetic. And a comparison on a common benchmark—a capsule skip entry reconstructed from published data—against Taylor-model flowpipes and against sampled trajectories of the full model, reported for each coordinate group rather than as a single number.

Success. Containment of every sample by the certified set, agreement with the cross-check where both apply, and a stated tightness against the baseline, coordinate by coordinate.

4 Plan of work

Six concrete obligations carry the program; they are stated where each arises above and gathered here, in order, as the plan of work.

1. *Interval embedding, uniform through the singular limit*: the outer approximation for the set-valued (interval) dynamics, with constants independent of ε .
2. *Validated computational pipeline*: interval evaluation, contraction and integration that discharge the certificate with every rounding error accounted for.
3. *Enclosures of the pass maps*: tight enclosures carried through a sequence of atmospheric passes without premature blow-up.
4. *Almost-orthogonality under interval disturbances*: non-interaction of the scaling groups under bounded uncertainty—**the critical path**, on which the profile decomposition and the closed-form bound rest.
5. *Closed-form constants*: $C_{\mathcal{R}}$ and K_{δ} from the physical corridor and the interval widths, free of any numerical solver.
6. *Independent verification*: symbolic checks of the derivations, a moment-sum-of-squares cross-check, and a benchmark against set-propagation tools and sampled trajectories.

Objectives 1–2 establish the interval occupation-measure embedding, 3–4 the profile decomposition, and 5–6 reduce it to closed form and check it; objective 4 gates the rest.

4.1 Where the work stands

The computational layers of the program have working prototypes; the central theorem is open. An implementation of the coupled model is public, with a suite of verification tasks whose failure criteria are stated before they run.¹ Three of its results bear directly on the claims above. The residual scaling has been measured rather than assumed: with the scale height varied over more than a decade, the distance to the glide scales as $\varepsilon^{0.49}$, and the closed form Eq. (2.4) bounds it from above to within about four percent. The translational equations of motion, the energy balance behind Eq. (2.2) and the linearization behind $C_{\mathcal{R}}$ are reduced to identities by computer algebra, against the same definition of the dynamics that the code evaluates in floating point and in interval arithmetic, and compiled numerical routines are generated from those expressions. And a validated-arithmetic prototype exists: enclosures of the entry field and reachable tubes in ball arithmetic, with a certified bound on the downrange still available to a capsule on a published skip entry, holding for every bank history that keeps the trajectory inside a stated corridor. None of this is the theorem. It is evidence that the constants are computable and that the pipeline can be built, and the same suite records the obstacle of Section 2.4, the weakly damped phugoid, as a computed fact rather than a conjecture.

4.2 Sequencing

The dependency structure suggests three phases rather than six parallel tracks. The first establishes the embedding—objective 1, together with the certified envelope of the atmospheric density that it needs in order to be stated on a concrete inclusion. The second attacks the critical path: objective 4,

¹<https://github.com/ethank5149/aether>

which is where the program either works or does not, and which should therefore be reached early enough that a negative result is still recoverable. Objective 3 runs alongside it, since the almost-orthogonality argument and the enclosure geometry constrain each other. The third phase reduces the decomposition to closed form—objective 5—completes the validated pipeline of objective 2, and runs the symbolic and moment-sum-of-squares checks of objective 6 against the result. The pipeline is the only deliverable that cannot be shortened by a weaker theorem.

Two retreat positions are worth identifying now. If objective 4 resists a proof for all three scaling groups simultaneously, the fallback is to establish it for the atmospheric and attitude groups and to treat the elastic group as a widened interval rather than a separate profile. That weakens the residual constant without invalidating the decomposition. And if the ringing left by a pass cannot be bounded in closed form, the phugoid amplitude is carried as one more slow coordinate of the skeleton, which costs a dimension and keeps the bound.

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